BUSINESS REQUIREMENTS DOCUMENT (BRD)

Meta Ad Performance Analysis

Business Objective

The business needs a performance tracking report for advertising campaigns running on Facebook and Instagram.
The report will provide visibility into campaign reach, engagement, conversions, and budget utilization.
This will enable the marketing team to:

- Identify the most effective platform (Facebook vs Instagram)
- Track campaign ROI and optimize budget allocation
- Understand audience engagement patterns

Scope of the Report

In Scope:
Campaigns running on Facebook and Instagram only.

Out of Scope:

- Other platforms (Messenger, Audience Network)
- Organic engagement (only paid ads will be included)

KPIs & Definitions

KPI	Definition	Formula	Example Use
Impressions	Number of times ads were displayed	Count of event_type = Impression	Measure reach
Clicks	Number of times users clicked ads	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared	Count of event_type = Share	Viral engagement
Comments	Number of user comments on ads	Count of event_type = Comment	User sentiment & feedback
Purchases	Number of purchases made after seeing ads	Count of event_type = Purchase	Conversions
Engagements	Total interactions (Clicks + Shares + Comments)	Clicks + Shares + Comments	Engagement volume

KPI	Definition	Formula	Example Use
CTR (Click Through Rate)	% of impressions that resulted in clicks	(Clicks ÷ Impressions) × 100	Ad effectiveness
Engagement Rate	% of impressions that resulted in engagements	(Engagements ÷ Impressions) × 100	Overall ad appeal
Conversion Rate	% of clicks that resulted in purchases	(Purchases ÷ Clicks) × 100	Funnel efficiency
Purchase Rate	% of impressions that resulted in purchases	(Purchases ÷ Impressions) × 100	Conversion from reach
Total Budget	Total spend allocated to campaigns	Sum of campaigns.total_budget	Cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign	Total Budget ÷ Campaign Count	Budget distribution

Visualizations/Charts Requirements:

1. Target Gender – Donut Chart

A donut chart will visualize performance split by target gender (from the ads table).
The metric displayed (e.g., Impressions, Clicks, Purchases) will change dynamically via a parameter.
Purpose: Identify which gender segment contributes most to the selected metric.

2. Target Age Group – Bar Chart

A bar chart will show engagement across age groups defined in the ads table.
Each bar represents one age group. The metric displayed will switch dynamically.
Purpose: Highlight which age group is most responsive to campaigns.

3. Country – Map

A map visualization will display performance by country (from the users table).
Bubble size or color intensity will represent the selected metric.
Purpose: Provide a geographic view of campaign reach and engagement.

4. Calendar Month – Calendar Heat Map

A calendar heat map will plot performance at the monthly level, based on the timestamp field in ad_events.
Darker shades indicate higher activity.
Purpose: Detect seasonal trends, peak ad months, and low-activity periods.

5. Weekly Trend – Stacked Column by Ad Type

A stacked column chart will display weekly performance trends.

- X-axis → Week number (from the Date Table linked to ad_events)
- Stacks → Different ad_type values (from the ads table)
- Y-axis → Selected metric
  Purpose: Compare ad type contributions over weeks.

6. Hourly Trend – Area Chart

An area chart will show activity by hour of day (from ad_events[time_of_day]).

- X-axis → Hour of the day (0–23)
- Y-axis → Selected metric
  Purpose: Understand user activity patterns throughout the day.

7. Ad Type – Matrix

A matrix visualization will show the selected metric across ad types and possibly break down further by platform (Facebook vs Instagram).

- Rows → Ad Types
- Columns → Platforms or other campaign dimensions
- Values → Selected metric
  Purpose: Compare performance across ad formats and platforms side by side.